

ValueRank

Ranking Information by Goal-Projected Change in User State

A two-page research proposal | Revised draft (v4) for academic discussion | July 2026

The **value** of a piece of information should be measured as the projection of a predicted, outcome-anchored change in the user's latent state onto a goal direction the user has *explicitly declared* — making the user's stated goal the **unit of value**, rather than a filter or re-ranking signal layered on top of engagement.

1. Problem & motivation

Modern ranking systems optimize *engagement* — a proxy that systematically diverges from whether information is genuinely valuable. Long-term-value (LTV) RL recommenders reduce myopia but (i) still measure value in *engagement units*, (ii) *infer* objectives from behavior rather than letting users *declare* them, and (iii) acquire documented incentives to *shift* user preferences toward whatever is easiest to satisfy [2]. Carroll et al. [11] make the structural problem precise: in **Dynamic Reward MDPs (DR-MDPs)**, where the reward function changes and is influenceable by the agent, alignment itself is ill-posed — *the reference against which influence should be judged is left undetermined*, and all eight alignment notions they compare either permit undesirable influence or over-constrain. A 2024–26 wave of *steerable* systems (natural-language recommender control [4], NL-profile steerability evaluation [12], value-based re-ranking) restores user agency — but these steer *which items surface* while the underlying value estimate remains an approval prediction. None of them changes *what value means*. We propose a constructive answer to the reference-indeterminacy that [11] exposes: **let the user declare the reference direction, and rank items by the predicted change they cause in the user, measured along that declared goal.**

2. Core idea

Represent three objects in a shared latent space: a user-profile state P_t (knowledge, skills, taste); a user-declared **intent direction** u ; and a candidate item x . Learn a transition model f predicting the state after consuming x , anchored to a measurable outcome. Define value as the **goal-projected state change**:

$$V(x | P_t, u) = \langle f(P_t, x) - P_t, A u \rangle$$

where A is a learned bilinear form correcting for anisotropy of the latent space ($A = I / \|u\|$ as the ablation). Rank candidates by V ; because the profile evolves across a session, selection maximizes *cumulative* goal-aligned change, $\sum_t \gamma^t V(x_t | P_t, u)$.

Theoretical grounding — user-side successor features, stated honestly. This decomposition is deliberately *structurally* isomorphic to the successor-features (SF) framework, where value factorizes as $\langle \psi, w \rangle$ and adapting to a new task vector requires only an inner product, with generalized policy improvement (GPI) providing transfer guarantees [1]. ValueRank transplants this structure across the system-user boundary: the “environment” whose features accumulate is the **user's own state**, and the task vector $w \equiv u$ is **set by the user, not the designer**. Steerability (rotating u re-orientes the ranking with no retraining) is a structural inheritance. **The guarantees are not.** SF/GPI assumes a fixed MDP, whereas the user's dynamics here are *endogenously non-stationary* — the system changes the very transition function it models, by design. In DR-MDP terms [11], the declared u pins the reference direction that [11] shows is otherwise undetermined; whether GPI-style transfer survives this endogeneity is an *open question we state as RQ2b rather than assume*. v3 of this proposal claimed the guarantees transfer; we retract that claim.

Intent elicitation. u is not a free latent direction, and not a revealed preference: following the argument that preferences alone are an inadequate unit for alignment [8], u is an *explicitly declared normative commitment*. Users state goals in natural language; an encoder maps the statement onto a low-dimensional, human-auditable *declared basis* (in the primary testbed: knowledge-component axes). The user sees and can edit the resulting basis weights. Non-identifiability is handled by construction, and the NL→ u mapping is itself an evaluated component, not an assumption.

3. Positioning — the precise gap

Line of work	What it does	What it does not do
LTV-RL recommenders PrefRec, ResAct [7]	Optimize discounted future engagement.	Value remains in engagement units; objective inferred, not declared.
NL-controllable rec. CTRL-Rec [4]; NL-profile steering & SteerEval [12]; value-toggle re-rankers	User's stated request re-weights predicted <i>approval</i> of items; steerability now has its own evaluation frameworks.	Value = predicted approval, not predicted user-state change ; steering targets item sets, not induced change; no shift audit.
Successor features / GPI Barreto et al. [1]	Value = $\langle \text{features}, \text{task vector} \rangle$; zero-shot task transfer <i>in a fixed MDP</i> .	Features describe the <i>environment</i> ; task vector set by designer; guarantees assume stationarity; never applied to user-state ranking.
Educational path RL ALPN [5]; CIKM-23 / KDD-24 [6]	Scalar learning-gain reward toward a fixed goal, education-only.	No vector intent, no inference-time steering, no domain-general operator.

Preference-change alignment Carroll et al. [2,3,11]	Diagnose / penalize induced shifts [2,3]; DR-MDPs formalize influence incentives and show 8 alignment notions all fail or over-constrain [11].	Negative / diagnostic results; no constructive value operator; the reference direction against which influence is judged is left undetermined.
Constrained ranking Brantley et al. [9]	Enforce long-term exposure / merit constraints on rankings.	Constraints are system-level and item-side; no user-declared objective, no user-state model.

The unoccupied gap

A value operator that is simultaneously: **(a) prospective and causal in the user** — scoring items by predicted change in an outcome-anchored user state, not by predicted approval; **(b) user-task SF** — the task vector is declared by the user at inference time, with GPI-style re-orientation; and **(c) shift-auditable** — because value is a projection, the induced state change decomposes exactly into a goal-aligned component $\Delta P_{\parallel}$ and an off-goal component $\Delta P_{\perp}$, with $\|\Delta P_{\perp}\|$ reported as a first-class safety metric. Read against [11]: the declared u is a candidate resolution of DR-MDP reference indeterminacy, and the audit makes the resulting influence measurable rather than assumed away. No prior system occupies the intersection; the nearest neighbors (CTRL-Rec on steerability, DR-MDP on the problem statement) each share one axis.

*Honest scope note: we do not claim to be “shift-transparent by construction.” Projection makes shift auditable **along and orthogonal to the declared axes**; shifts outside the declared basis, and manipulation of the profile estimate rather than the user, remain live risks and are treated as such (§7, RQ4).*

4. Formalization

- **State model.** $P_t \in \mathbb{R}^d$, grounded in a measurable downstream outcome (held-out quiz/recall in the primary testbed) so f cannot collapse to the trivial “no-change” solution and profile-estimate gaming is detectable against ground truth. The basis dimension d is a declared design parameter: auditability requires small, human-readable d , at a stated cost in expressiveness (§7).
- **Transition model.** $P_{t+1} = f_{\theta}(P_t, x_t)$, trained on outcome-anchored next states, with an ensemble / epistemic-uncertainty head; drift of f across a session is logged as the empirical non-stationarity measure used in RQ2b.
- **Policy.** π maximizes $\sum \gamma^t [V(x_t) - \lambda \cdot \sigma_f(x_t)]$, where σ_f is model uncertainty. The penalty is *required, not optional*: because V is computed from our own model, an unpenalized policy will exploit f 's errors (classical model-based-RL hacking). Periodic ground-truth probes re-anchor f during training, not only at evaluation.
- **Baseline hierarchy.** Greedy-V is the myopic baseline; RL is justified *only* if long-horizon planning beats greedy — pre-registered as a genuine possible negative result.

5. Research questions (with pre-specified metrics)

- **RQ1 (validity — the decisive comparison).** Does the learned projection operator V predict held-out outcome gain better than (i) engagement signals, (ii) LLM-predicted approval, and — critically — (iii) a **goal-conditioned LLM re-ranker** prompted directly with the user's stated goal? Baseline (iii) is the head-to-head that decides whether the machinery is justified at all: *if a prompted LLM matches V , ValueRank reduces to prompt engineering and we say so.* Metric: outcome-gain regression / ranking AUC vs. all three.
- **RQ2 (steerability & linearity).** (a) When u rotates by $\Delta\theta$, does the top-k re-order predictably? Metrics: rank correlation between $\Delta\theta$ and top-k Kendall- τ displacement; goodness-of-fit of the linear reward assumption (SF error-bound term), reported explicitly. (b) **RQ2b — non-stationarity:** how fast does the SF-style decomposition degrade as f 's dynamics drift within a session, and do empirical drift-rate bounds exist under which GPI-style re-orientation still holds? This operationalizes the open question of §2.
- **RQ3 (horizon).** Does RL beat greedy-V on cumulative goal attainment, and under what curriculum-dependency structure? Metric: items-to-mastery; outcome gain at fixed budget.
- **RQ4 (shift & gaming audit).** Is induced shift better aligned with declared intent than an LTV baseline, and how large is the off-goal component? Metrics: $\|\Delta P_{\parallel}\| / \|\Delta P_{\perp}\|$ ratio; $\|\Delta P_{\perp}\|$ vs. a no-system natural-drift estimate; profile-estimate vs. ground-truth divergence. The gaming detector must itself resist *teaching-to-the-test*: probes are item-blind and drawn from a generator held out from the policy, and we test directly whether the item-selection policy predicts the probe distribution (it must not).

6. Evaluation plan

Primary testbed — language learning. Profile = knowledge state from knowledge tracing; u = target-proficiency direction over knowledge components; ground truth = held-out recall/quiz. Non-circular anchor; connects to learning-gain baselines. Residual circularity risk acknowledged: here $V \approx$ learning gain nearly by construction — which is precisely why the secondary testbed carries the domain-general claim.

Secondary testbed — topic mastery in news/technical reading, run to the same standard. Profile = concept-familiarity state probed by short comprehension quizzes; items = articles. v3 called this “lightweight”; *we withdraw that hedge*. It is pre-registered with the same metrics as the primary testbed, because it carries the domain-general claim: if ValueRank only wins where goal-projected gain $\approx$ learning gain, the claim contracts — in print — to “an education-domain ranking operator.” A simulated-user environment (LLM-based, pre-specified fidelity-validation protocol [10]) supports the RL experiments; if fidelity validation fails, RQ3 is reported as *unanswered* rather than answered simulator-only.

Baselines. Engagement ranking; contextual bandit; LTV-RL; spaced-repetition scheduler; ALPN-style scalar-gain policy; CTRL-Rec-style stated-preference re-weighting; constrained-ranking control [9]; and the goal-conditioned LLM re-ranker, promoted to the decisive RQ1 comparison above.

Ablations. Remove intent projection ($\rightarrow$ pure learning gain); fix u ($\rightarrow$ non-steerable); remove RL ($\rightarrow$ greedy); $A = I$ ($\rightarrow$ raw inner product); remove uncertainty penalty ($\rightarrow$ measure reward hacking directly).

7. Risks & honest limits

- **Guarantee non-transfer (sharpest theoretical risk).** SF/GPI guarantees assume a fixed MDP; our environment is a user whose dynamics the system changes by design. The isomorphism may be *structural only*. RQ2b measures the degradation empirically; deriving drift conditions under which transfer survives — or a proof that it cannot — is the priority theory target. Until then, steerability is an engineered property with an SF-shaped motivation, not a guaranteed one.
- **Off-basis shift.** The audit covers the declared basis and its orthogonal complement within the learned space; changes the space cannot represent are invisible. Mitigation: outcome anchoring plus periodic basis-expansion checks. A limit, stated as one.
- **Teaching to the test.** A policy could inflate probe performance without goal-general learning if item selection predicts probe content. Mitigation: item-blind randomized probes from a held-out generator, plus the RQ4 predictability test. *Residual risk:* a sufficiently expressive policy may still exploit shared structure between items and any probe family; we report this exposure rather than claim it closed.
- **Auditability-expressiveness tradeoff.** The audit is only meaningful if the declared basis is small and human-readable; small d limits what goals can be expressed. d is therefore reported alongside every result, and claims are scoped to goals representable in the declared basis.
- **Linearity.** If goal progress is strongly non-linear in the latent space, the SF-style decomposition degrades; RQ2 quantifies exactly how much, and the bilinear A is the first-line fix.
- **Greedy suffices.** Pre-registered possible outcome; the operator V remains a contribution even without the RL layer.
- **Goal quality — a limitation, not an extension.** Real users hold vague, conflicting, self-undermining goals; single- u is a scoping choice that excludes them and bounds the external validity of every result here. A weighted multi-goal cone is future work; paternalism (should the system second-guess a declared goal?) is out of scope — per the indeterminacy analysis of [11], there may be no neutral answer.
- Beyond the two testbeds, ground-truth value stays hard to measure; “taste as an asset” remains discussion-only.

8. Anticipated contributions

1. A **constructive counterpart to the DR-MDP negative result [11]**: fixing the reference direction by *user declaration at inference time*, yielding **user-side successor features** — the SF/GPI decomposition transplanted to the system-user boundary as a new unit of ranking value, with its guarantee conditions stated as open rather than assumed.
2. An **outcome-anchored user-transition model** with an explicit anti-hacking (uncertainty-penalized, re-anchored, probe-blind) training loop.
3. A **shift-audit protocol** ($\Delta P_{\parallel}/\Delta P_{\perp}$ decomposition + profile-gaming detector) as a constructive complement to post-hoc preference-shift diagnosis [2,3].
4. Empirical evidence on **when sequential planning beats greedy goal-projection** — in either direction — and on whether a learned projection operator beats a prompted LLM re-ranker at all.

Selected related work

[1] Barreto, A., et al. (2017). Successor Features for Transfer in Reinforcement Learning. NeurIPS; and Barreto et al. (2018), ICML (SF&GPI with reward-features).
 [2] Carroll, M., Dragan, A., Russell, S., & Hadfield-Menell, D. (2022). Estimating and Penalizing Induced Preference Shifts in Recommender Systems. ICML. arXiv:2204.11966.
 [3] Carroll, M., Chan, A., Ashton, H., & Krueger, D. (2023). Characterizing Manipulation from AI Systems. EAAMO. arXiv:2303.09387.
 [4] Carroll, M., Foote, A., Feng, K., Williams, M., Dragan, A., Knox, W.B., & Milli, S. (2025). CTRL-Rec: Controlling Recommender Systems With Natural Language. arXiv:2510.12742.

[5] Chen, J.-Y., Saeedvand, S., & Lai, I.-W. (2023). Adaptive Learning Path Navigation Based on Knowledge Tracing and RL. arXiv:2305.04475.
 [6] Li, Q., et al. (2023). Graph Enhanced Hierarchical RL for Goal-oriented Learning Path Recommendation. CIKM; and Li et al. (2024), Privileged Knowledge State Distillation, KDD.
 [7] Xue, W., et al. (2023). PrefRec. KDD; and Xue et al. (2022), ResAct. arXiv:2206.02620.
 [8] Zhi-Xuan, T., Carroll, M., Franklin, M., & Ashton, H. (2024). Beyond Preferences in AI Alignment. Philosophical Studies.

[9] Brantley, K., Fang, Z., Dean, S., & Joachims, T. (2024). Ranking with Long-Term Constraints. WSDM.
 [10] Zhang, Z., et al. (2025). LLM-Powered User Simulator for Recommender System. AAAI.
 [11] Carroll, M., Foote, D., Siththaranjan, A., Russell, S., & Dragan, A. (2024). AI Alignment with Changing and Influenceable Reward Functions. ICML. arXiv:2405.17713.
 [12] SteerEval: A Framework for Evaluating Steerability with Natural Language Profiles for Recommendation (2026). arXiv:2601.21105.

Note to reader. Three sharp claims: (i) ValueRank is a **constructive response to the DR-MDP reference-indeterminacy result [11]** — the user declares the direction alignment is measured against; (ii) the SF framing is **structural**; its guarantees under endogenous non-stationarity are an open question, not an assumption (changed from v3); (iii) the projection yields a **shift audit, not shift immunity**. The decisive test is the goal-conditioned LLM re-ranker (RQ1). Feedback on guarantee transfer, the teaching-to-the-test exposure, and the bilinear form A is especially welcome.